

LESSON 4-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Estimate with Benchmarks**

Maryland Standard 6.RP.A.3c • Learning target – I can estimate a percent by comparing an amount to benchmark percents.

REMEMBER

- **Benchmarks:** 10%, 25%, 50%, 75%, 100%.
- Find benchmark amounts of the **whole**. Which one is the part **about** equal to?
- A **double number line** lines up amounts with percents.

WORKED EXAMPLE

A water jug holds 40 cups. There are 29 cups left. About what percent of the jug is left?

Step 1 Benchmarks of 40: $50\% = 20$ and $75\% = 30$.

Step 2 29 is very close to 30.

Step 3 So 29 out of 40 is about 75%.

Answer about 75%

WORD BANK

benchmark – an easy percent: 10%, 25%, 50%, 75%

the whole – the full amount; it is 100%

about – close to, not exact

estimate – a close answer

PRACTICE

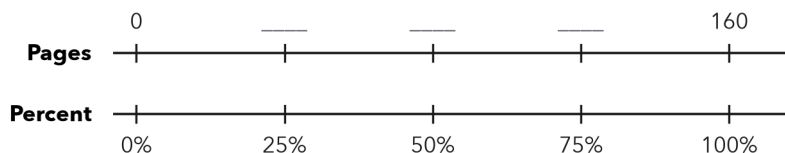
show your work • label units

- 1. Reading Goal.** Jada's goal is to read 160 pages. She has read 42 pages. Fill in the double number line. Then estimate the percent of her goal she has read.

Step 1 Find $160 \div 4$. That is one jump on the top line.

Step 2 Fill in each jump on the top line.

Step 3 Find the mark closest to 42.



About _____ %

- 2.** About what percent? Name the closest benchmark: 10%, 25%, 50%, 75%, or 100%.

a. 7 out of 60 $\approx$ ____ % **b.** 38 out of 50 $\approx$ ____ % **c.** 23 out of 90 $\approx$ ____ %

d. 19 out of 40 $\approx$ ____ % **e.** 49 out of 50 $\approx$ ____ % **f.** 11 out of 45 $\approx$ ____ %

g. 16 out of 150 $\approx$ ____ % **h.** 61 out of 80 $\approx$ ____ %

- 3. Class Fundraiser.** A class wants \$500. They have **raised** \$96 so far. Use 10% of \$500 to estimate the percent they have raised.

Step 1 Find 10%: $\$500 \div 10$.

Step 2 Double it to get 20%. Is \$96 close?

a. 10% of \$500 = _____

b. About what percent have they raised? _____ %

- 4. Bus Seats.** A bus has 48 seats. 35 seats are filled. About what percent of the seats are filled? Name the benchmark you used.

Step 1 Find 50% and 75% of 48.

Step 2 Which amount is 35 closest to?

About _____ %

- 5. Compare.** Which is closer to 50%: 26 out of 50, or 26 out of 60? Explain how you know.

Frame ____ out of ____ is closer to 50% because half of 50 is ____ and half of 60 is ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Look at a battery icon or a gas gauge. Estimate the percent it shows and explain your benchmark to someone at home.

LESSON 4-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Estimate a Percent of a Number**

Maryland Standard 6.RP.A.3c • Learning target – I can estimate a percent of a number using friendly percents and rounded amounts.

REMEMBER

- Round to **friendly numbers** first.
- 10%: divide by 10. 5% = half of 10%.
- 25% = $1/4$, 50% = $1/2$, 75% = $3/4$.

WORKED EXAMPLE

Estimate 24% of 83.

Step 1 Round: 25% of 80.

Step 2 25% = $1/4$, and $80 \div 4 = 20$.

Answer about 20

WORD BANK

estimate – a close answer, not exact

round – change to a friendly number

10% – the whole $\div 10$

friendly number – easy to use, like 40 or 200

PRACTICE

show your work • label units

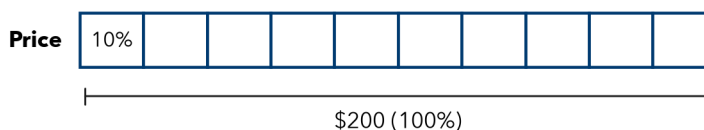
1. Estimate. Round to friendly numbers first.

Step 1 Round the percent and the number.

Step 2 Find 10%, then build the percent you need.

- a. 11% of 59 $\approx$ ____ b. 74% of 41 $\approx$ ____ c. 26% of 203 $\approx$ ____

2. **Sneaker Sale.** Sneakers cost \$198 and are 30% off. Round to \$200. Each part of the tape is 10% of \$200.



a. About how many dollars is one part (10%)? _____

b. About how much is the discount (30%)? _____

3. **School Lunch.** 412 students. About 48% **buy lunch**. About how many students buy lunch?

Step 1 Round: 48% is about 50%. 412 is about 400.

Step 2 50% is half. Find half of 400.

About _____ students

4. **Apple Orchard.** A farm picked 798 apples. About 24% of them were too small to sell. About how many apples were too small?

Step 1 Round: 24% is about 25%. 798 is about 800.

Step 2 25% is one fourth. Divide by 4.

About _____ apples

- 5. Find the Error.** Mia estimated 19% of 512 by writing $20 \times 500 = 10,000$. Explain her mistake. Give a good estimate.

Frame Mia's mistake is _____. 20% means _____, so a good estimate is _____.

TEST PRACTICE

choose the best answer

- 6.** Which is the best estimate of 26% of 79?

A 2 **B** 20 **C** 26 **D** 40

- 7.** A coat costs \$119. It is 40% off. Which is the best estimate of the discount?

Step 1 Round \$119 to \$120.

Step 2 Find 10% of \$120, then multiply by 4.

A \$4.80 **B** \$40 **C** \$48 **D** \$72

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Find a sale sign. Estimate the discount out loud with someone at home.